10th Grade
Term 1
C5 Practice Activity

Evaluated Criterion

C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss.

1 Minimax Regret Criterion

When probabilities are not available, a decision-maker can compare alternatives by considering the opportunity loss, or regret, associated with each possible state of nature. Regret measures the payoff that could have been obtained by choosing the best alternative for the state of nature that actually occurs, minus the payoff obtained from the selected alternative.

For a payoff table with alternatives in rows and states of nature in columns, let $P_{ay}(A_i, S_j)$ represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

For each state of nature, first identify the best available payoff:

$$\begin{aligned} b(S_j) &= \max_i P_{ay}(A_i, S_j) \\ &= \max\left\{P_{ay}(A_1, S_j), P_{ay}(A_2, S_j), \ldots, P_{ay}(A_m, S_j)\right\}. \end{aligned}$$

The regret associated with alternative A_i under state S_j is then

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For each alternative, determine its maximum possible regret:

$$\begin{aligned} R(A_i) &= \max_j r(A_i, S_j) = \max_j \left\{b(S_j) - P_{ay}(A_i, S_j)\right\} \\ &= \max\left\{r(A_i, S_1), r(A_i, S_2), \ldots, r(A_i, S_n)\right\}. \end{aligned}$$

The Minimax Regret criterion selects the alternative whose maximum regret is as small as possible:

$$A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i).$$

Thus, the procedure is:

> Best payoff in each state $\longrightarrow$ Regret table $\longrightarrow$ Maximum regret for each alternative
> Minimum of those maximum regrets

The selected alternative minimizes the largest opportunity loss that could result from the decision.

2 General 2x2 Problem

Consider the following payoff table:

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Construct the opportunity loss table and determine the decision using the Minimax Regret criterion.

2.1 Solution

Step 1: Identify the best payoff in each state of nature

Under S_1, compare the payoffs of A_1 and A_2:

$$b(S_1) = \max\{P_{ay}(A_1, S_1), P_{ay}(A_2, S_1)\}.$$

Under S_2, compare the payoffs of A_1 and A_2

$$b(S_2) = \max\{P_{ay}(A_1, S_2), P_{ay}(A_2, S_2)\}.$$

Therefore,

$$b(S_1) = \max\{P_{ay}(A_1, S_1), P_{ay}(A_2, S_1)\},$$
$$b(S_2) = \max\{P_{ay}(A_1, S_2), P_{ay}(A_2, S_2)\}.$$

Step 2: Construct the opportunity loss table

For each alternative and state of nature, subtract the payoff from the best payoff available under that state:

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

Thus,

$$r(A_1, S_1) = b(S_1) - P_{ay}(A_1, S_1),$$

$$r(A_1, S_2) = b(S_2) - P_{ay}(A_1, S_2),$$

$$r(A_2, S_1) = b(S_1) - P_{ay}(A_2, S_1),$$

$$r(A_2, S_2) = b(S_2) - P_{ay}(A_2, S_2).$$

Step 3: Find the maximum regret for each alternative

$$\text{For} A_1, \qquad R(A_1) = \max\{r(A_1, S_1), r(A_1, S_2)\}.$$

For A_2,

$$R(A_2) = \max\{r(A_2, S_1), r(A_2, S_2)\}.$$

Therefore,

$$R(A_1) = \max\{b(S_1) - P_{ay}(A_1, S_1), b(S_2) - P_{ay}(A_1, S_2)\},$$
$$R(A_2) = \max\{b(S_1) - P_{ay}(A_2, S_1), b(S_2) - P_{ay}(A_2, S_2)\}.$$

Step 4: Apply the Minimax Regret criterion

Compare the maximum regrets:

$$R^* = \min\{R(A_1), R(A_2)\}.$$

The Minimax Regret decision is

$$A^*_{\text{Minimax Regret}} = \arg\min_i \{R(A_i)\}.$$

Thus, choose A_1 if $R(A_1) < R(A_2)$, choose A_2 if $R(A_2) < R(A_1)$, and report a tie if $R(A_1) = R(A_2)$.

Opportunity loss table

Alternative	S_1	S_2	Maximum Regret
A_1	$r(A_1, S_1)$	$r(A_1, S_2)$	$R(A_1)$
A_2	$r(A_2, S_1)$	$r(A_2, S_2)$	$R(A_2)$

3 Contextualized 2x2 Problem - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager knows that the level of customer traffic during the promotional weekend is uncertain and has no reliable information about the probability of high or low traffic.

What decision should the manager make using the Minimax Regret criterion?

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

3.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Launch dessert line, A_2 = Keep current menu States of nature: S_1 = High traffic, S_2 = Low traffic.

Step 2: Identify the best payoff in each state of nature

Under high traffic, compare the payoffs of both alternatives:

$$b(S_1) = \max\{42, 28\} = 42.$$

Under low traffic:

$$b(S_2) = \max\{8, 20\} = 20.$$

Thus, the best payoff in each state is

$$b(S_1) = 42,$$
$$b(S_2) = 20.$$

Step 3: Construct the opportunity loss table

Regret is calculated as

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1:

$$r(A_1, S_1) = 42 - 42 = 0,$$

$$r(A_1, S_2) = 20 - 8 = 12.$$

For A_2:

$$r(A_2, S_1) = 42 - 28 = 14,$$

$$r(A_2, S_2) = 20 - 20 = 0.$$

Therefore, the opportunity loss table is

	S_1	S_2
A_1	0	12
A_2	14	0

Step 4: Find the maximum regret for each alternative

For A_1:

$$R(A_1) = \max\{0, 12\} = 12.$$

For A_2:

$$R(A_2) = \max\{14, 0\} = 14.$$

Therefore,

$$R(A_1) = 12,$$
$$R(A_2) = 14.$$

Step 5: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{12, 14\} = 12.$$

Thus,

$$A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i) = A_1.$$

The Minimax Regret criterion recommends launching the dessert line because its maximum possible regret is \$12,000, compared with \$14,000 for keeping the current menu.

Extended opportunity loss table

Alternative	S_1	S_2	Maximum Regret
A_1: Launch dessert line	0	12	$\max\{0, 12\} = 12$
A_2: Keep current menu	14	0	$\max\{14, 0\} = 14$

4 Contextualized 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager knows that customer response to the campaign is uncertain and has no reliable information about the probability of strong or weak sales.

What decision should the manager make using the Minimax Regret criterion?

Alternative	Strong sales (S_1)	Weak sales (S_2)
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

4.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Launch new campaign, A_2 = Keep current strategy States of nature: S_1 = Strong sales, S_2 = Weak sales.

Step 2: Identify the best payoff in each state of nature

Under strong sales, compare the payoffs of both alternatives:

$$b(S_1) = \max\{55, 35\} = 55.$$

Under weak sales:

$$b(S_2) = \max\{5, 22\} = 22.$$

Thus,

$$b(S_1) = 55,$$
$$b(S_2) = 22.$$

Step 3: Construct the opportunity loss table

Regret is calculated as

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1:

$$r(A_1, S_1) = 55 - 55 = 0,$$

$$r(A_1, S_2) = 22 - 5 = 17.$$

For A_2:

$$r(A_2, S_1) = 55 - 35 = 20,$$

$$r(A_2, S_2) = 22 - 22 = 0.$$

Therefore, the opportunity loss table is

	S_1	S_2
A_1	0	17
A_2	20	0

Extended opportunity loss table

Step 4: Find the maximum regret for each alternative

For A_1:

$$R(A_1) = \max\{0, 17\} = 17.$$

For A_2:

$$R(A_2) = \max\{20, 0\} = 20.$$

Therefore,

$$R(A_1) = 17,$$
$$R(A_2) = 20.$$

Step 5: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{17, 20\} = 17.$$

Thus,

$$A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i) = A_1.$$

The Minimax Regret criterion recommends launching the new campaign because its maximum possible regret is \$17,000, compared with \$20,000 for keeping the current strategy.

Alternative	S_1	S_2	Maximum Regret
A_1: Launch new campaign	0	17	$\max\{0, 17\} = 17$
A_2: Keep current strategy	20	0	$\max\{20, 0\} = 20$

5 Contextualized 4x3 Problem - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager knows that market conditions are uncertain and has no reliable information about the probability of each possible level of demand. The company is considering four possible strategies: launching a premium version, launching a standard version, launching a budget version, or delaying the launch.

The level of market demand could be high, medium, or low. The estimated profit for each strategy under each possible market condition is shown in the table below.

What decision should the company make using the Minimax Regret criterion?

Alternative	High demand (S_1)	Medium demand (S_2)	Low demand (S_3)
Premium version (A_1)	80	35	-20
Standard version (A_2)	60	42	18
Budget version (A_3)	40	32	25
Delay launch (A_4)	15	15	15

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

5.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Premium version, A_2 = Standard version, A_3 = Budget version, A_4 = Delay launch

States of nature: S_1 = High demand, S_2 = Medium demand, S_3 = Low demand.

Step 2: Identify the best payoff in each state of nature

Under high demand, compare the payoffs of all alternatives:

$$b(S_1) = \max\{80, 60, 40, 15\} = 80.$$

Under medium demand:

$$b(S_2) = \max\{35, 42, 32, 15\} = 42.$$

Under low demand:

$$b(S_3) = \max\{-20, 18, 25, 15\} = 25.$$

Thus,

$$b(S_1) = 80,$$
$$b(S_2) = 42,$$
$$b(S_3) = 25.$$

Step 3: Construct the opportunity loss table

Regret is calculated as

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1:

$$r(A_1, S_1) = 80 - 80 = 0,$$
$$r(A_1, S_2) = 42 - 35 = 7,$$
$$r(A_1, S_3) = 25 - (-20) = 45.$$

For A_2:

$$r(A_2, S_1) = 80 - 60 = 20,$$
$$r(A_2, S_2) = 42 - 42 = 0,$$
$$r(A_2, S_3) = 25 - 18 = 7.$$

For A_3:

$$r(A_3, S_1) = 80 - 40 = 40,$$
$$r(A_3, S_2) = 42 - 32 = 10,$$
$$r(A_3, S_3) = 25 - 25 = 0.$$

For A_4:

$$r(A_4, S_1) = 80 - 15 = 65,$$
$$r(A_4, S_2) = 42 - 15 = 27,$$
$$r(A_4, S_3) = 25 - 15 = 10.$$

Therefore, the opportunity loss table is

	S_1	S_2	S_3
A_1	0	7	45
A_2	20	0	7
A_3	40	10	0
A_4	65	27	10

Step 4: Find the maximum regret for each alternative

For A_1:

$$R(A_1) = \max\{0, 7, 45\} = 45.$$

For A_2:

$$R(A_2) = \max\{20, 0, 7\} = 20.$$

For A_3:

$$R(A_3) = \max\{40, 10, 0\} = 40.$$

For A_4:

$$R(A_4) = \max\{65, 27, 10\} = 65.$$

Therefore,

$$R(A_1) = 45,$$
$$R(A_2) = 20,$$
$$R(A_3) = 40,$$
$$R(A_4) = 65.$$

Step 5: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{45, 20, 40, 65\} = 20.$$

Thus,

$$A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i) = A_2.$$

The Minimax Regret criterion recommends launching the standard version because its maximum possible regret is \$20,000, which is lower than the maximum regret of every other strategy.

Extended opportunity loss table

Alternative	S_1	S_2	S_3	Maximum Regret
A_1: Premium version	0	7	45	$\max\{0, 7, 45\} = 45$
A_2: Standard version	20	0	7	$\max\{20, 0, 7\} = 20$
A_3: Budget version	40	10	0	$\max\{40, 10, 0\} = 40$
A_4: Delay launch	65	27	10	$\max\{65, 27, 10\} = 65$